TEXORA · LECTURE SHEET

Hosiery Machines

A hosiery machine is a small circular machine — 3 to 5 inches of cylinder — that knits a complete shaped tube: welt, leg, heel pocket, foot, and a toe that modern machines close automatically. Shaping here is done in the round, by knitting on part of the needles while the rest hold their loops.

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1. Machine types

TYPE	CONSTRUCTION	PRODUCT
Single-cylinder	One cylinder of latch needles + sinkers; rib welt by dial-less means (ribber attachments on some)	Plain socks, sports socks (with terry sinkers)
Cylinder + dial (rib) hosiery	Small dial above cylinder, rib gating	True rib-top socks, school socks
Fine hosiery machines	High needle counts (up to ~400 needles), very fine gauges, sheer yarns	Stockings, tights/pantyhose (knitted as two legs joined, or seamless)

Needle count fixes sock size range together with yarn and stitch length: e.g. a 4-inch cylinder at E28-equivalent spacing carries $N = \pi \times 4 \times 28 \approx$ **352 needles** in the ideal count; commercial machines quote their own needle numbers (96–200 for men's sports socks, up to 400 for fine hosiery) — always work from the machine's stated needle count in calculations.

2. Anatomy of a sock, knitted in order

1. **Welt/cuff:** rib or mock-rib (tuck-based) with elastane inlay for grip; a doubled welt folds over on some styles.
2. **Leg:** plain, rib, or patterned (selection) courses; terry loops inside for sports socks (plush sinkers).
3. **Heel pocket:** the shaping step (§3).
4. **Foot:** plain courses; terry sole on sports styles; reinforced heel/toe yarns where specified.
5. **Toe:** a second short-row pocket (on many machines) or a wedge, then the opening is closed — traditionally by linking/looping off-machine, on modern machines by an automatic closing device that sews the toe course-by-course on the machine.

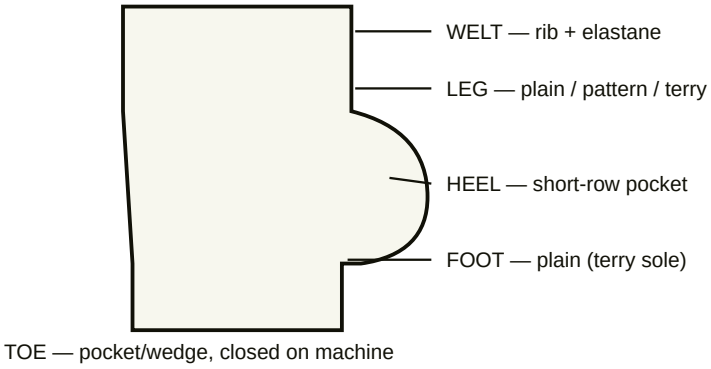


Figure 1 — The sock as knitted: one continuous tube with two shaped pockets. Nothing is cut; fit comes from structure, elastane and the heel/toe shaping.

3. Heel formation by short-row (reciprocating) knitting

For the heel, the machine stops rotating continuously and **reciprocates** (oscillates) over roughly half the cylinder — the heel needles — while the instep half holds its loops idle. In successive part-courses, needles at the edges of the working group are taken out of action one (or two) at a time (**narrowing**), so the knitted pouch deepens in the middle; then the needles are returned one by one (**widening**) until the full circle knits again. The result is a three-dimensional pocket at right angles to the tube — the same narrowing/widening arithmetic as flat fashioning (Topic 10), executed in the round. **Worked check:** heel group = 176 needles (half of 352); narrowing removes 1 needle per side per part-course for 22 part-courses → working needles fall by 44 to 132 at the pouch centre; widening restores them over the next 22 part-courses. Total heel part-courses ≈ 44 plus the turning courses — which is why heel time is a visible share of the sock cycle.

4. Cycle time and production

WORKED EXAMPLE — SOCKS PER SHIFT

A machine knits a sports sock in 3.2 minutes of pure knitting; add 8 % of cycle time for doffing/handling losses on top of knitting (efficiency applied as: productive minutes per 8-h shift = 480 × 0.92 = 441.6 min).

Socks per shift = 441.6 ÷ 3.2 = **138 socks = 69 pairs** (5.75 dozen pairs) per machine-shift — always report pairs, not single socks, to buyers.

A 100-machine hall at the same rate: 6900 pairs per shift. Small cycle-time savings (0.2 min) add ≈ 8 socks per machine-shift — the reason heel/toe automation and fast closing matter commercially.

Yarns: cotton and cotton blends for sports socks, nylon/polyester for durability and fine hosiery, elastane in welt and arch bands. Quality checks: size (foot length on a form), heel symmetry, toe-seam

quality, terry uniformity, elastane grin and course count per size programme.

Glossary

Reciprocating — oscillating knitting over part of the cylinder for heel/toe. **Short-row** — a part-course knitted on a subset of needles. **Mock rib** — rib appearance from tuck selection on one cylinder. **Linking** — joining toe loops loop-for-loop (hand or machine) for a flat seam. **Plush/terry sock** — sinker-loop pile on the inside.

Key points

• Hosiery = small circular machines knitting a complete shaped article: welt → leg → heel pocket → foot → toe (auto-closed on modern frames).
• Heel/toe shaping is narrowing then widening over a reciprocating half-cylinder group.
• Production is counted in cycle minutes per sock — verify pairs vs singles in every report.

Practice

1. A sock cycle is 2.8 min; shift efficiency 90 % over 480 min. Find socks and pairs per shift.
2. A heel group has 120 needles; narrowing takes out 2 needles per side per part-course for 15 part-courses. How many needles work at the pouch centre?
3. Why is the welt knitted with elastane inlay rather than relying on rib contraction alone?
4. State two quality differences between a linked toe and an auto-closed toe as felt by the wearer, if any, and the production difference.

Answer key

1. Productive minutes = 480 × 0.90 = 432; socks = 432 ÷ 2.8 = **154 socks = 77 pairs** per shift.
2. Removed = 2 sides × 2 × 15 = 60; working = 120 − 60 = **60 needles**.
3. Rib contraction alone loosens permanently with wear and washing (loop relaxation); elastane's continuous retractive force maintains grip over the garment's life and across size tolerance.
4. Wearer: a well-linked toe is the flattest; a good auto-close is nearly as flat — both beat a coarse overlocked seam ridge. Production: linking is a separate off-machine operation with labour and handling; auto-closing completes the sock on the machine, removing that operation.